# Muaz Khalifa Al Radi

**Ph.D. Candidate in Electrical and Computer Engineering**

Khalifa University, Abu Dhabi, UAE

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| **Phone:** | +971 56 902 3971 |
| **Google Scholar:** | Google Scholar Profile |
| **Scopus:** | 57217631941 |
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| **LinkedIn:** | Profile |
| **GitHub:** | MR81224 |
| **Languages:** | Fluent in spoken and written Arabic and English |


*Research areas:* AI-Enabled Robotic Perception · Autonomous UAV Inspection & Visual Servoing · Video Anomaly Detection · Vision-Language Models & Multimodal AI

## Profile

My research develops AI-enabled visual perception systems for autonomous inspection and anomaly detection, combining deep learning, vision-language models, and visual servoing control to enable reliable operation in safety-critical and industrially challenging environments. With over 1,500 Google Scholar citations and a growing publication record across robotics, computer vision, image processing, and AI-driven engineering systems, I bring research depth and five years of hands-on teaching experience. My goal is to build an independent research program at the intersection of autonomous visual intelligence, multimodal AI, and trustworthy perception for real-world infrastructure and safety systems.

## Research Interests

Computer Vision · Autonomous Robotic Inspection · UAV Perception & Visual Servoing · Video Anomaly Detection · Vision-Language Models · Weakly Supervised Learning · Foundation Models for Robotic Perception · Industrial AI & Safety Monitoring

## Future Research Program

- Developing unified foundation model frameworks that combine visual and language reasoning for generalizable robotic inspection across diverse industrial assets.
- Designing label-efficient anomaly detection systems that operate under weak, noisy, or open-set supervision for deployment in safety-critical surveillance infrastructure.
- Building semantics-aware UAV navigation systems capable of task-driven scene understanding and adaptive path planning in unstructured environments.
- Establishing evaluation benchmarks and explainability tools for trustworthy AI perception in high-stakes domains such as energy facilities, oil & gas, and environmental monitoring.
- Investigating cross-modal robustness and domain adaptation of perception models under adverse real-world conditions including smoke, heat haze, glare, and occlusion.

## Academic Qualifications

**Ph.D. in Electrical and Computer Engineering** — **Expected: Dec. 2026**

Khalifa University, Abu Dhabi, UAE — Current CGPA: 3.88 / 4.00

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| *Thesis:* | "Learning to Detect Anomalies in Video Surveillance" |
| *Supervisor:* | Dr. Sajid Javed |
| *Committee:* | Prof. Naoufel Werghi; Prof. Jorge Manuel Miranda Dias; Dr. Abdulhadi Shoufan |
| *Defense:* | October 2026 |
| *Abstract:* | Investigates multimodal video anomaly detection by integrating visual features with vision-language and large language models. The work explores reduced-supervision strategies—including weak and few-shot labelling—to build scalable, generalizable detection systems for real-world surveillance. |

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**M.Sc. in Electrical and Computer Engineering** **2022**

Khalifa University, Abu Dhabi, UAE CGPA: 4.00 / 4.00

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| *Thesis:* | "Autonomous Inspection of Flare Stacks Using a Visual Servoing Controlled Unmanned Aerial Vehicle (UAV)" |
| *Supervisor:* | Prof. Jorge Manuel Miranda Dias |
| *Abstract:* | Developed a UAV-based robotic inspection framework for flare stack monitoring using image-based visual servoing and deep learning-based visual analytics. The system was validated end-to-end in simulation using the AirSim environment. |

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**B.Sc. in Sustainable and Renewable Energy Engineering** **2020**

University of Sharjah, UAE CGPA: 3.97 / 4.00 — Highest Honors, Ranked 1st

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| *Project:* | "Design of CubeSat Power System" |
| *Supervisor:* | Dr. Chaouki Ghenai |
| *Abstract:* | Designed a solar-powered electrical system for a ship-tracking CubeSat, optimising component selection and orbital parameters through a parametric study. System performance was simulated in MATLAB and STK, with parameter optimisation carried out using Design Expert software. |

## Professional Experience

**Graduate Research and Teaching Assistant (GRTA)** **Jan. 2021 – Present**

Visual Signal Analysis and Processing (VSAP) Group, Khalifa University Center for Advanced Robotic Systems (KU-CARS), Khalifa University, Abu Dhabi, UAE

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| *Research focus:* | Robotic inspection, computer vision, deep learning, and visual servoing control. |
| *Key contributions:* | Developed vision-based UAV inspection methods for flare stack operation monitoring; implemented image-based visual servoing (IBVS) for autonomous UAV data collection; designed deep learning pipelines for flare stack detection and smoke analysis; contributed to multimodal LLM/VLM-based anomaly assessment methods accepted at *IEEE ICIP 2024*. |
| *Teaching:* | Four undergraduate computing courses over 5+ years (details in Teaching Experience section). |

**Research Assistant** **Sep. 2018 – Jan. 2021**

Research Institute for Science and Engineering (RISE), University of Sharjah, UAE

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| *Supervisor:* | Prof. Mohammad Ali Abdulkareem |
| *Research topics:* | Capacitive Deionization (CDI) desalination, renewable energy-powered desalination, AI applications in renewable energy, heat pipes. |
| *Key skills:* | Electrochemical testing (Gamry, CHI workstations); CV, GCD, and EIS characterisation; AutoCAD cell design for CDI applications; engineering team supervision; review paper preparation. |

## Teaching Experience

**Graduate Teaching Assistant** **Jan. 2021 – Present**

Department of Computer Science, Khalifa University, Abu Dhabi, UAE

Served as teaching assistant, lab instructor, and project mentor for four undergraduate computing courses over five years, accumulating ten semesters of experience across lecturing and lab instruction. Each lab session opened with a short lecture on the topic before transitioning to the hands-on exercise. Responsibilities included preparing course and lab content, designing and grading lab assessments, mentoring students individually, and overseeing the preparation and delivery of quizzes and course projects.

**COSC 202 — Data Science and Artificial Intelligence** *5 semesters*

Comprehensive survey of data science and AI with engineering applications. Topics include data analysis and visualisation, machine learning, deep learning, computer vision, natural language processing, autonomous driving, and AI ethics. Students gain hands-on experience with Python-based AI and data science libraries.

**ENGR 114 — Introduction to Computing: Python** *2 semesters*

Foundations of programming and computational problem-solving using Python. Topics include data types, control flow, functions, data structures (lists, tuples, databases), and file I/O.

**ENGR 113 — Introduction to Computing: MATLAB** *2 semesters*

Programming principles and computational workflows using MATLAB. Topics include arrays, functions, scripts, logical operators, and introductory data analysis and input/output operations.

**ENGR 112 — Introduction to Computing: C++** *1 semester*

Introduction to structured programming using C++. Topics include data types, conditional and repetitive structures, functions, arrays, strings, and file handling.

**Instructor — AI Winter Camp 2025** **Dec. 15–19, 2025**

Khalifa University (Outreach Department) in partnership with the Abu Dhabi Department of Education and Knowledge (ADEK), Abu Dhabi, UAE

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| *Theme:* | "AI for a Sustainable Future" |
| *Participants:* | 63 high school students, Grades 10–12, selected through an academic interview process |
| *Format:* | Five-day in-person program of lectures, hands-on workshops, and a culminating Grand Challenge project |
| *Role:* | Delivered lectures and supervised hands-on sessions throughout the program |
| *Program:* | *Day 1:* ML fundamentals, Python environment setup, dataset import, model training and evaluation, OpenCV image capture. *Day 2:* Data quality and its impact on ML model performance; hands-on data cleaning exercises. *Day 3:* Comparative study of ML models; introduction of the Grand Challenge dataset. *Day 4:* Grand Challenge completion — students trained models and integrated them into applications; judging and evaluation. *Day 5:* AI ethics; winner announcement; certificate distribution. |

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**Online Educator — SREE Tutorials (YouTube Channel)** **Ongoing**

Arabic-language engineering education channel delivering university-level lecture content to a global audience.

**137** videos **6,170** subscribers **651,680** total views

*Courses:* Deep Learning System Design, Fluid Mechanics, Heat Transfer, Thermodynamics, Engineering Materials, Feedback Control Systems, Applied Electronics, Electric Power Engineering, Wind Energy Systems, Solar PV Systems.

## Curriculum Development and Accreditation Support

**Accreditation Evidence Preparation for Undergraduate Computing Courses** **Jan. 2021 – Present**

Department of Computer Science, Khalifa University, Abu Dhabi, UAE

- Contributed to curriculum documentation and accreditation evidence preparation for undergraduate computing courses in support of Commission for Academic Accreditation (CAA) and ABET Engineering Accreditation Commission (EAC) reporting processes.
- Prepared and organized course assessment packages at the end of each semester, including lab manuals, quizzes, course projects, unsolved assessment versions, solution keys, and anonymized graded samples.
- Compiled low-, mid-, and high-performance student samples for lab assessments, quizzes, and course projects, ensuring that grading, feedback, and justification were clearly documented while preserving student anonymity.
- Supported quality assurance of course delivery by maintaining structured evidence of student learning, assessment alignment, and grading consistency across computing laboratory courses.

## Courses Prepared to Teach

### Undergraduate Courses

- Programming for Engineers: Python, MATLAB, and C++
- Introduction to Artificial Intelligence
- Data Science for Engineers
- Machine Learning and Deep Learning
- Computer Vision
- Robotics and Autonomous Systems
- Control Systems

- Signals and Systems
- Renewable Energy Systems

**Graduate Courses**

- Deep Learning for Computer Vision
- Autonomous Robotic Perception
- Vision-Language Models and Multimodal AI
- Video Understanding and Anomaly Detection
- AI for Industrial Inspection and Safety Monitoring

## Student Supervision and Mentorship

**Capstone Project Mentor — Design of Flare Stack System for Autonomous UAV Inspection** **Khalifa University**

- Mentored a senior design/capstone team on the design, development, and testing of a flare stack simulator for autonomous UAV inspection experiments.
- Guided students in system design, component selection, experimental setup, safety considerations, and testing of a functional flaring experimental system suitable for micro-UAV inspection research.
- The student project was later built upon by the KU–ADNOC flare stack research group and supported subsequent research activities in vision-based flare analytics.
- Related research output: Boumaraf, S., **Al Radi, M.**, Abdelhafez, F.O., Li, P., Al Awadhi, K.Y., Karki, H., Dhelim, S., & Werghi, N. (2025). Vision-based air-flow monitoring in an industrial flare system design using deep convolutional neural networks. *Expert Systems with Applications*, 272, 126733.

**Freshman Engineering Project Mentor** **University of Sharjah**

- Mentored freshman engineering student teams in the Department of Sustainable and Renewable Energy Engineering for the course "0406100 Introduction to Energy Science and Technology."
- Supervised hands-on renewable energy projects designed to demonstrate students' understanding of fundamental renewable energy applications.
- Projects mentored across three semesters included: PV Cooler; Water Distillation Using a Solar Still; and Solar-Powered Remote-Controlled Cleaning Car.

**Undergraduate Course Project and Laboratory Mentorship** **Khalifa University**

- Mentored undergraduate students during computing laboratory sessions in Python, MATLAB, C++, Data Science, and Artificial Intelligence.
- Guided student teams in hands-on AI and data science course projects involving real-world problem formulation, data understanding, exploratory data analysis, preprocessing, machine learning model development, hyperparameter tuning, evaluation, and deployment on unseen data.

**AI Winter Camp Project Supervision** **Khalifa University**

- Supervised and assessed high-school student teams during the AI Winter Camp Grand Challenge, where students developed AI-based applications using concepts learned during the program.
- Provided project guidance on dataset preparation, model training, evaluation, and practical integration of AI models into student-built applications.

## Selected Publications

*Selected publications emphasizing first-author and major-contribution work in computer vision, autonomous robotic inspection, video anomaly detection, vision-language models, and AI-enabled engineering systems.*

1. **[Q1 Journal, Top 7%, First Author] Al Radi, M.** & Javed, S. (2026). Hierarchical vision-language model with comprehensive language description for video anomaly detection. *Knowledge-Based Systems*, 115466.

2. **[CVPR 2026]** Alawode, B., Mahmood, A., **Al Radi, M.K.**, Albastaki, S., Khan, A., Bilal, M., Abdalla, M.A., Bennamoun, M., & Javed, S. (2026). MLLM-HWSI: A multimodal large language model for hierarchical whole slide image understanding. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 13732–13743.
3. **[Q1 Journal, Top 9%, First Author] Al Radi, M.**, Li, P., Boumaraf, S., Dias, J., Werghi, N., Karki, H., & Javed, S. (2024). AI-enhanced gas flares remote sensing and visual inspection: Trends and challenges. *IEEE Access*, 12, 56249–56274.
4. **[Q1 Journal, Top 7%, First Author] Al Radi, M.**, AlMallahi, M.N., Al-Sumaiti, A.S., Semeraro, C., Abdelkareem, M.A., & Olabi, A.G. (2024). Progress in artificial intelligence-based visual servoing of autonomous unmanned aerial vehicles (UAVs). *International Journal of Thermofluids*, 21, 100590.
5. **[Q1 Journal, Top 10%]** Abdalla, M., Javed, S., **Al Radi, M.**, Ulhaq, A., & Werghi, N. (2025). Video anomaly detection in 10 years: A survey and outlook. *Neural Computing and Applications*, 37(32), 26321–26364.
6. **[ICIP 2024, Co-First Author]** Li, P., **Al Radi, M.**, Elmezain, M.S., Ahmed, A.H., Boudiaf, A., Boumaraf, S., Dias, J., Karki, H., Javed, S., & Al Awadhi, K.Y. (2024). SMO-CLIP: Enhancing anomalous smoke density assessment using a hybrid LLM-VLM approach. *2024 IEEE International Conference on Image Processing (ICIP)*, 760–765.
7. **[ICAR 2023, First Author] Al Radi, M.**, Boumaraf, S., Karki, H., Dias, J., Werghi, N., & Javed, S. (2023). Vision-based analytics of flare stacks using deep learning detection. *2023 21st International Conference on Advanced Robotics (ICAR)*, 467–472. IEEE.
8. **[IECON 2023, First Author] Al Radi, M.**, Li, P., Karki, H., Werghi, N., Javed, S., & Dias, J. (2023). Multi-view inspection of flare stacks operation using a vision-controlled autonomous UAV. *IECON 2023 – 49th Annual Conference of the IEEE Industrial Electronics Society*, 1–6. IEEE.
9. **[Book Chapter, First Author] Al Radi, M.**, Karki, H., Werghi, N., Javed, S., & Dias, J. (2023). Autonomous inspection of flare stacks using an unmanned aerial system. In *Unmanned Aerial Vehicles Applications: Challenges and Trends* (pp. 201–223). Springer.

## Publications

*Google Scholar (as of June 2026): Citations **1,564**, h-index **18**, i10-index **23**.*
*Scopus (as of June 2026): Citations **1,233**, h-index **15**, Documents **36**.*

### Book Chapters

1. **Al Radi, M.**, Karki, H., Werghi, N., Javed, S., & Dias, J. (2023). Autonomous inspection of flare stacks using an unmanned aerial system. In *Unmanned Aerial Vehicles Applications: Challenges and Trends* (pp. 201–223). Springer.

### Journal Papers

1. **Al Radi, M.** & Javed, S. (2026). Hierarchical vision-language model with comprehensive language description for video anomaly detection. *Knowledge-Based Systems*, 115466.
2. Abdalla, M., Javed, S., **Al Radi, M.**, Ulhaq, A., & Werghi, N. (2025). Video anomaly detection in 10 years: A survey and outlook. *Neural Computing and Applications*, 37(32), 26321–26364.
3. Boumaraf, S., **Al Radi, M.**, Abdelhafez, F.O., Li, P., Al Awadhi, K.Y., Karki, H., Dhelim, S., & Werghi, N. (2025). Vision-based air-flow monitoring in an industrial flare system design using deep convolutional neural networks. *Expert Systems with Applications*, 272, 126733.
4. Boumaraf, S., Li, P., **Al Radi, M.**, Abdelhafez, F.O., Behouch, A., Al Awadhi, K.Y., Karki, H., Javed, S., Dias, J., & Werghi, N. (2025). Optimized flare performance analysis through multi-modal machine learning and temporal standard deviation enhancements. *IEEE Access*.
5. **Al Radi, M.**, Li, P., Boumaraf, S., Dias, J., Werghi, N., Karki, H., & Javed, S. (2024). AI-enhanced gas flares remote sensing and visual inspection: Trends and challenges. *IEEE Access*, 12, 56249–56274.

6. **Al Radi, M.**, AlMallahi, M.N., Al-Sumaiti, A.S., Semeraro, C., Abdelkareem, M.A., & Olabi, A.G. (2024). Progress in artificial intelligence-based visual servoing of autonomous unmanned aerial vehicles (UAVs). *International Journal of Thermofluids*, 21, 100590.
7. Abdelkareem, M.A., **Al Radi, M.**, Mahmoud, M., Sayed, E.T., Salameh, T., Alqadi, R., Kais, E.C.A., & Olabi, A.G. (2024). Recent progress in wind energy-powered desalination. *Thermal Science and Engineering Progress*, 47, 102286.
8. Olabi, A.G., Abdelghafar, A.A., Soudan, B., Alami, A.H., Semeraro, C., **Al Radi, M.**, Al-Murisi, M., & Abdelkareem, M.A. (2024). Artificial neural network driven prognosis and estimation of lithium-ion battery states: Current insights and future perspectives. *Ain Shams Engineering Journal*, 15(2), 102429.
9. Sayed, E.T., Obaid, M., Olabi, A.G., Abdelkareem, M.A., **Al Radi, M.**, Al-Dawoud, A., Al-Ashei, S., & Ghaffour, N. (2023). Recent progress on the application of capacitive deionization for wastewater treatment. *Journal of Water Process Engineering*, 56, 104379.
10. Sayed, E.T., Olabi, A.G., Elsaid, K., **Al Radi, M.**, Semeraro, C., Doranehgard, M.H., Eltayeb, M.E., & Abdelkareem, M.A. (2023). Application of artificial intelligence techniques for modeling, optimizing, and controlling desalination systems powered by renewable energy resources. *Journal of Cleaner Production*, 413, 137486.
11. Sayed, E.T., Olabi, A.G., Shehata, N., **Al Radi, M.**, Muhaisen, O.M., Rodriguez, C., Atieh, M.A., & Abdelkareem, M.A. (2023). Application of bio-based electrodes in emerging capacitive deionization technology for desalination and wastewater treatment. *Ain Shams Engineering Journal*, 14(8), 102030.
12. Sayed, E.T., Olabi, A.G., Elsaid, K., **Al Radi, M.**, Alqadi, R., & Abdelkareem, M.A. (2023). Recent progress in renewable energy based-desalination in the Middle East and North Africa (MENA) region. *Journal of Advanced Research*, 48, 125–156.
13. **Al Radi, M.** & Ghenai, C. (2023). Artificial neural network PV performance prediction and electric power system simulation of a ship-tracking CubeSat. *Iranian Journal of Science and Technology, Transactions of Electrical Engineering.*
14. Olabi, A.G., Abdelghafar, A.A., Maghrabie, H.M., Sayed, E.T., Rezk, H., **Al Radi, M.**, Obaideen, K., & Abdelkareem, M.A. (2023). Application of artificial intelligence for prediction, optimization, and control of thermal energy storage systems. *Thermal Science and Engineering Progress*, 39, 101730.
15. Semeraro, C., Olabi, A.G., Aljaghoub, H., Alami, A.H., **Al Radi, M.**, Dassisti, M., & Abdelkareem, M.A. (2023). Digital twin application in energy storage: Trends and challenges. *Journal of Energy Storage*, 58, 106347.
16. Olabi, A.G., Haridy, S., Sayed, E.T., **Al Radi, M.**, Alami, A.H., Zwayyed, F., Salameh, T., & Abdelkareem, M.A. (2023). Implementation of artificial intelligence in modeling and control of heat pipes: A review. *Energies*, 16(2), 760.
17. Olabi, A.G., Abdelkareem, M.A., Semeraro, C., **Al Radi, M.**, Rezk, H., Muhaisen, O., Al-Isawi, O.A., & Sayed, E.T (2023). Artificial neural networks applications in partially shaded PV systems. *Thermal Science and Engineering Progress*, 37, 101612.
18. Abdelkareem, M.A., Soudan, B., Mahmoud, M.S., Sayed, E.T., AlMallahi, M.N., Inayat, A., **Al Radi, M.**, & Olabi, A.G. (2022). Progress of artificial neural networks applications in hydrogen production. *Chemical Engineering Research and Design*, 182, 66–86.
19. **Al Radi, M.**, Adil Al-Isawi, O., Abdelghafar, A.A., Fayez Abu Qiyas, A., AlMallahi, M., Khanafer, K., & Assad, M.E.H. (2022). Recent progress, economic potential, and environmental benefits of mineral recovery geothermal brine treatment systems. *Arabian Journal of Geosciences*, 15(9), 832.
20. Abdelkareem, M.A., Maghrabie, H.M., Sayed, E.T., Kais, E.C.A., Abo-Khalil, A.G., **Al Radi, M.**, Baroutaji, A., & Olabi, A.G. (2022). Heat pipe-based waste heat recovery systems: Background and applications. *Thermal Science and Engineering Progress*, 29, 101221.
21. Maghrabie, H.M., Olabi, A.G., Alami, A.H., **Al Radi, M.**, Zwayyed, F., Wilberforce, T., & Abdelkareem, M.A. (2022). Numerical simulation of heat pipes in different applications. *International Journal of Thermofluids*, 16, 100199.

22. **Al Radi, M.**, Sayed, E.T., Alawadhi, H., & Abdelkareem, M.A. (2021). Progress in energy recovery and graphene usage in capacitive deionization. *Critical Reviews in Environmental Science and Technology*, 52(17), 3080–3136.
23. Sayed, E.T., **Al Radi, M.**, Ahmad, A., Abdelkareem, M.A., Alawadhi, H., Atieh, M.A., & Olabi, A.G. (2021). Faradic capacitive deionization (FCDI) for desalination and ion removal from wastewater. *Chemosphere*, 275, 130001.

### Conference Papers

1. Alawode, B., Mahmood, A., **Al Radi, M.K.**, Albastaki, S., Khan, A., Bilal, M., Abdalla, M.A., Bennamoun, M., & Javed, S. (2026). MLLM-HWSI: A multimodal large language model for hierarchical whole slide image understanding. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 13732–13743.
2. **Al Radi, M.**, Ahmad, A., & Boudiaf, A. (2025). Robotic optimal path planning via multi-objective dolphin echolocation optimization (DEO). *2025 Advances in Science and Engineering Technology International Conferences (ASET)*, 1–6. IEEE.
3. Li, P., **Al Radi, M.**, Elmezain, M.S., Ahmed, A.H., Boudiaf, A., Boumaraf, S., Dias, J., Karki, H., Javed, S., & Al Awadhi, K.Y. (2024). SMO-CLIP: Enhancing anomalous smoke density assessment using a hybrid LLM-VLM approach. *2024 IEEE International Conference on Image Processing (ICIP)*, 760–765.
4. Ahmed, A., Velayudhan, D., ElMezain, M., **Al Radi, M.**, Boudiaf, A., Hassan, T., Deriche, M., Bennamoun, M., & Werghi, N. (2024). CLIFS: CLIP-driven few-shot learning for baggage threat classification. *2024 IEEE International Conference on Image Processing (ICIP)*, 753–759.
5. **Al Radi, M.**, Ahmad, A., Ahmad, A., Boudiaf, A., Alkaabi, N., Malouf, M., & Jelinek, H. (2024). Machine learning-based multivariate classification of cardiac autonomic neuropathy via imbalanced ECG recordings. *2024 Advances in Science and Engineering Technology International Conferences (ASET)*, 1–5. IEEE.
6. Li, P., **Al Radi, M.**, Huang, X., Boumaraf, S., Shen, Y., Guo, F., Al Awadhi, K.Y., Dias, J., Javed, S., & Karki, H. (2024). ES-ATF: Early smoke detection based on attention-aggregated temporal feature extraction. *2024 International Conference on Engineering and Emerging Technologies (ICEET)*, 1–6. IEEE.
7. Al Radi, H., Al Salem, S.A., Rami, D., Gouda, A.M., Mustafa, A., Abdulrahman, F., Hanieh, M.A., Farid, M., **Al Radi, M.**, & Bani-Issa, W. (2024). Caffeine consumption and mental health correlations in healthcare students: A cross-sectional study and machine learning analysis. *2024 Advances in Science and Engineering Technology International Conferences (ASET)*, 1–6. IEEE.
8. Ahmad, A., **Al Radi, M.**, Ahmad, A., & Boudiaf, A. (2024). Reinforcement learning-based optimal control design for active suspension systems. *2024 Advances in Science and Engineering Technology International Conferences (ASET)*, 1–6. IEEE.
9. **Al Radi, M.**, Boumaraf, S., Karki, H., Dias, J., Werghi, N., & Javed, S. (2023). Vision-based analytics of flare stacks using deep learning detection. *2023 21st International Conference on Advanced Robotics (ICAR)*, 467–472. IEEE.
10. **Al Radi, M.**, Li, P., Karki, H., Werghi, N., Javed, S., & Dias, J. (2023). Multi-view inspection of flare stacks operation using a vision-controlled autonomous UAV. *IECON 2023 – 49th Annual Conference of the IEEE Industrial Electronics Society*, 1–6. IEEE.
11. Ahmed, A.H., **Al Radi, M.**, & Werghi, N. (2023). An ensemble learning method based on deep neural and PCA-based SVM network for baggage threat and smoke recognition. *2023 Advances in Science and Engineering Technology International Conference (ASET)*, 1–6. IEEE.
12. **Al Radi, M.**, Karki, H., Werghi, N., Javed, S., & Dias, J. (2022). Video analysis of flare stacks with an autonomous low-cost aerial system. *Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC 2022).* SPE.
13. **Al Radi, M.**, Karki, H., Werghi, N., Javed, S., & Dias, J. (2022). Vision-based inspection of flare stacks operation using a visual servoing controlled autonomous unmanned aerial vehicle (UAV). *IECON 2022 – 48th Annual Conference of the IEEE Industrial Electronics Society*, 1–6. IEEE.

14. **Al Radi, M.**, Ahmed, M.A., Hassan, M.M., & Ghenai, C. (2022). Performance analysis, parametric study, and optimization of CubeSat solar power system for ship tracking of marine traffic. *2022 Advances in Science and Engineering Technology International Conferences (ASET)*, 1–6. IEEE.
15. **Al Radi, M.**, Hassan, M.M., Ahmed, M.A., & Ghenai, C. (2020). Design of CubeSat solar power system for real-time tracking of Sharjah vessel. *2020 Advances in Science and Engineering Technology International Conferences (ASET)*, 1–5. IEEE.

## Conference Presentations

**IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026)** — **3–7 June 2026**
*Colorado Convention Center, Denver, Colorado, USA*

▹ Virtually presented "MLLM-HWSI: A Multimodal Large Language Model for Hierarchical Whole Slide Image Understanding."

**UAE Graduate Students Research Conference (GSRC 2026)** — **16 May 2026**
*University of Sharjah, Sharjah, UAE*

▹ Presented "LHTM-VAD: Language-Guided Hierarchical Temporal Modeling for Video Anomaly Detection."

**2025 Advances in Science and Engineering Technology International Conferences (ASET)** — **15–18 Sep. 2025**
*Dubai, UAE*

▹ Presented "Robotic Optimal Path Planning via Multi-Objective Dolphin Echolocation Optimization (DEO)."

**2024 IEEE International Conference on Image Processing (ICIP)** — **27–30 Oct. 2024**
*Abu Dhabi, UAE*

▹ Co-authored paper "SMO-CLIP: Enhancing Anomalous Smoke Density Assessment Using a Hybrid LLM-VLM Approach."

▹ Co-authored paper "CLIFS: CLIP-Driven Few-Shot Learning for Baggage Threat Classification."

**21st International Conference on Advanced Robotics (ICAR 2023)** — **5–8 Dec. 2023**
*Abu Dhabi, UAE*

▹ Presented "Vision-Based Analytics of Flare Stacks Using Deep Learning Detection."

**IECON 2023 — 49th Annual Conference of the IEEE Industrial Electronics Society** — **16–19 Oct. 2023**
*Singapore*

▹ Presented "Multi-View Inspection of Flare Stacks Operation Using a Vision-Controlled Autonomous UAV."

**2023 Advances in Science and Engineering Technology International Conferences (ASET)** — **20–23 Feb. 2023**
*Dubai, UAE*

▹ Presented "An Ensemble Learning Method Based on Deep Neural and PCA-Based SVM Network for Baggage Threat and Smoke Recognition."

**Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC 2022)** — **31 Oct.–3 Nov. 2022**
*Abu Dhabi, UAE*

▹ Presented "Video Analysis of Flare Stacks with an Autonomous Low-Cost Aerial System."

**IECON 2022 — 48th Annual Conference of the IEEE Industrial Electronics Society** — **18–21 Oct. 2022**
*Brussels, Belgium*

▹ Presented "Vision-Based Inspection of Flare Stacks Operation Using a Visual Servoing Controlled Autonomous UAV."

**UAE Graduate Students Research Conference (GSRC 2022)** — **24 Mar. 2022**
*Khalifa University, Abu Dhabi, UAE*

▹ Presented "Implementing Visual Servoing Control for Drone-Based Inspection of Flare Stack Structure."

**2022 Advances in Science and Engineering Technology International Conferences (ASET)** — **21–24 Feb. 2022**
*Dubai, UAE*

▹ Presented "Performance Analysis, Parametric Study, and Optimization of CubeSat Solar Power System for Ship Tracking of Marine Traffic."

**2020 Advances in Science and Engineering Technology International Conferences (ASET)** — **4–6 Feb. 2020**
*Dubai, UAE*

▹ Presented "Design of CubeSat Solar Power System for Real-Time Tracking of Sharjah Vessel."

**12th International Conference on Sustainable Energy & Environmental Protection (SEEP 2019)** — **18–21 Nov. 2019**
*University of Sharjah, Sharjah, UAE*

▹ Presented "Energy Recovery and Use of Graphene as Electrode Material for Capacitive Deionization: A State-of-the-Art Review."

## Funded and Industry-Collaborative Research Projects

**Vision-Based Flare Analytics** — **Khalifa University–ADNOC Research Collaboration**
*Role: Research Team Member*

- Contributed to an externally funded industry-collaborative research project focused on AI-enabled visual inspection and analytics for flare stack monitoring.
- Developed and supported research activities in UAV-based visual data collection, flare and smoke visual analytics, image-based visual servoing, and deep learning-based inspection pipelines.
- The project resulted in multiple peer-reviewed publications, conference presentations, and research outputs in autonomous inspection, computer vision, and industrial AI.
- The KU–ADNOC research team received the "Technical Innovation of the Year" award at the Oil and Gas Middle East Awards 2024 for the Vision-Based Flare Analytics project.

## Scholarships

**Combined M.Sc. & Ph.D. Research and Teaching Scholarship (CMDRTS)** — **Jan. 2021 – Dec. 2026**
*Khalifa University, Abu Dhabi, UAE — Full research and teaching scholarship covering tuition, monthly stipend, and conference funding for the combined M.Sc./Ph.D. program.*

**Academic Distinction Discount** — **Sep. 2016 – Jun. 2020**
*University of Sharjah, UAE — Awarded for maintaining a CGPA above 3.60/4.00 throughout the B.Sc. program.*

## Honors, Awards & Competitions

**Technical Innovation of the Year – Government Sector** — **Mar. 2024**
*Oil and Gas Middle East Awards 2024 — Awarded as a member of the Khalifa University–ADNOC research team for the project "Vision-Based Flare Analytics".*

**IES Students & Young Professionals Activity Contest Winner** — **Oct. 2022**
*IEEE Industrial Electronics Society (IES-SYPA), IECON 2022, Brussels, Belgium — Won for the paper "Vision-Based Inspection of Flare Stacks Using a Visual Servoing Controlled Autonomous UAV". Awarded up to $1,500 USD in travel funding.*

**UAE 10-Year Golden Visa** — **Sep. 2021 – Sep. 2031**
*UAE Government — Granted under the category "Outstanding Students and Graduates" for graduating with a CGPA of 3.97/4.00 and ranking first in the B.Sc. cohort.*

**Shortlisted Finalist, Future Generation Competition** — **Mar. 2020**
*Middle East Energy 2020, Dubai, UAE — Selected as a finalist in a regional competition showcasing outstanding engineering student projects across the MENA region. Presented the project "Design of CubeSat Power System" to industry professionals and conference attendees.*

**Certificates of Appreciation (3 awards)** **Jun. 2020**
*Department of Sustainable and Renewable Energy Engineering, University of Sharjah — Received for outstanding performance in the Senior Design Project, academic performance during 2019–2020, and logistics support to SREE students.*

**2nd Place, Engineering Science Clubs Exhibition** **Mar. 2019**
*3D Printers & Drones category, as a member of the SREE Club, University of Sharjah, UAE.*

## Peer Review Service

Reviewed manuscripts for peer-reviewed journals and international conferences in artificial intelligence, computer vision, robotics, measurement, and energy systems.

- *Engineering Applications of Artificial Intelligence*
- *Gas Science and Engineering*
- *Measurement*
- *Pattern Recognition*
- *Knowledge-Based Systems*
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

## Academic Service & Outreach

**Session Chair** **Mar. 2022**
*Session "EPS-B4: Robotics, Mechatronics & Automation", UAE Graduate Students Research Conference (GSRC 2022), Khalifa University, UAE.*

**Competition Judge** **Nov. 2019**
*Judging committee member, 12th Renewable Challenge, SREE Department, University of Sharjah, UAE.*

**Head of Tutorials Team** **Sep. 2018 – May. 2019**
*Led the tutorials team of the SREE Club, University of Sharjah; prepared and delivered educational content for undergraduate engineering students.*

**Student Club Organiser** **University of Sharjah**
*Core member of the SREE Club organising team; coordinated events, workshops, and competitions including the annual Renewable Challenge.*

## Technical Skills

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|---|---|
| **Programming** | Python, MATLAB, C++ |
| **Machine Learning, Deep Learning & Data Science** | PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, CUDA, Jupyter Notebook, Google Colab |
| **Computer Vision & Multimodal AI** | OpenCV, Hugging Face Transformers, vLLM |
| **Robotics & Simulation** | MATLAB/Simulink, AirSim, Unreal Engine 4, Pix4D |
| **Development Tools** | Linux, Git/GitHub, LaTeX |
| **Documentation & Productivity** | Microsoft Word, Excel, PowerPoint |